



FACULTY
OF COMPUTATIONAL
MATHEMATICS
AND CYBERNETICS



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GazeMorph: Reducing Eye-Tracking Measurement Error by Fixation-to-Content Alignment

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Eye-Tracking Measurement Error Problem

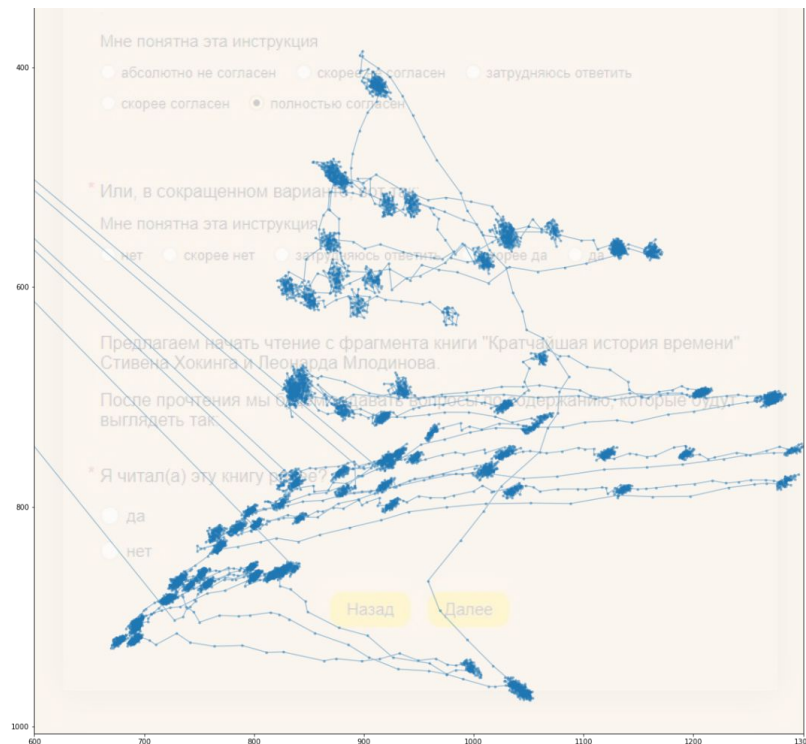
A major challenge in eye-tracking is spatial measurement error: recorded fixations may be shifted away from their true targets.

Measurement errors may include:

- constant offset
- nonlinear distortion
- scale mismatch

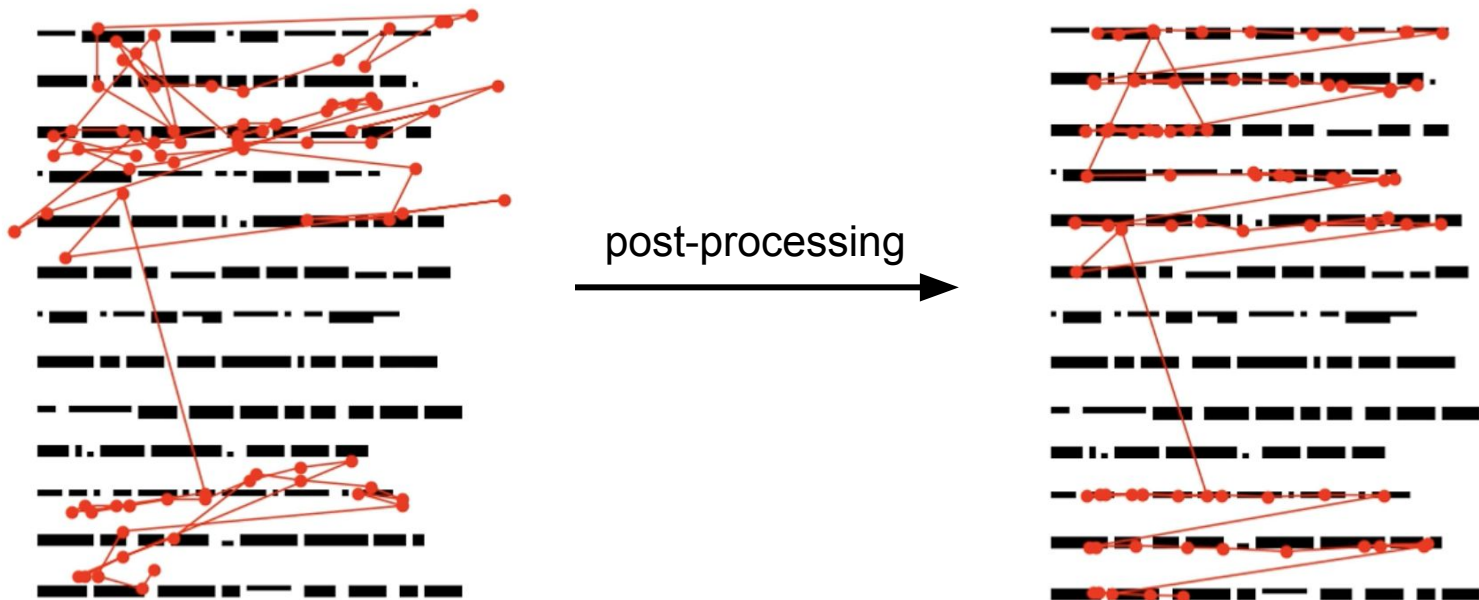
These errors can also drift over time due to head movement or changes in the recording environment, such as lighting.

Consequence: noisy gaze coordinates reduce the reliability of downstream analyses.

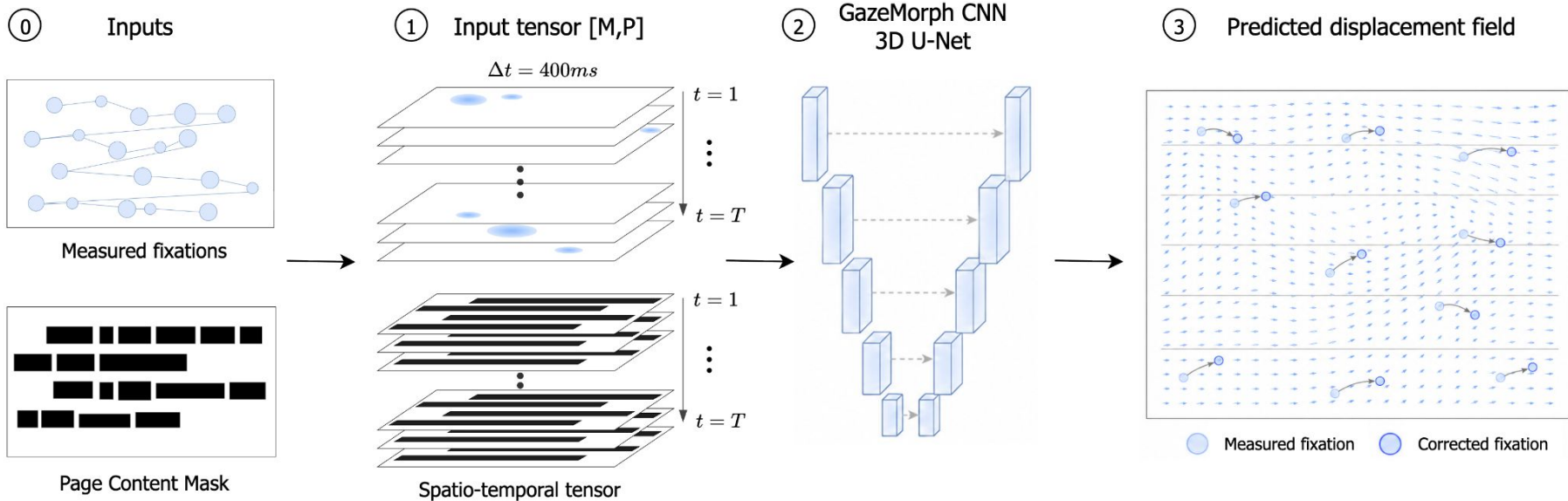


Correction as a Fixation-to-Content Alignment

People usually look at content, not background → gaze geometry tends to follow content geometry



Proposed approach



Corrected fixation = measured fixation + predicted displacement

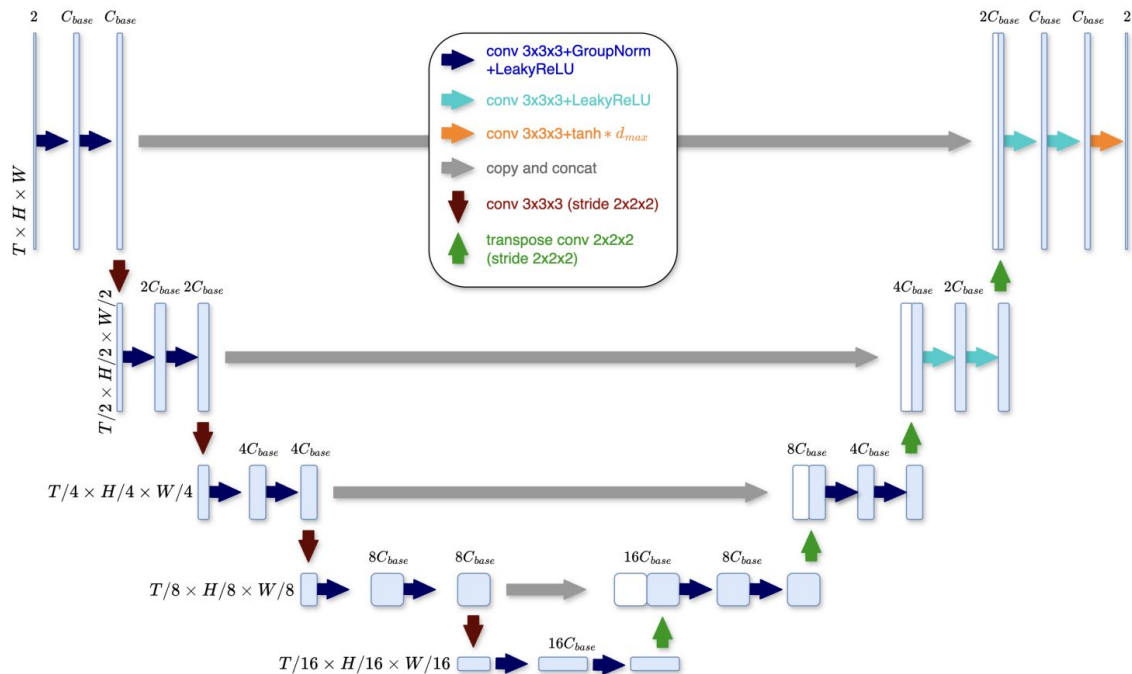
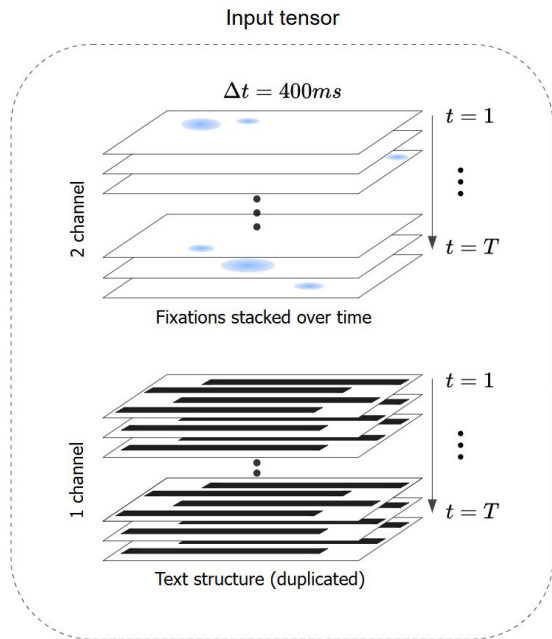
$$\hat{f}_i = m_i + U(t_i, x_i, y_i)$$

$\hat{f}_i = (\hat{t}_i^f, \hat{x}_i^f, \hat{y}_i^f)$ — corrected i -th fixation

$m_i = (t_i^m, x_i^m, y_i^m)$ — measured i -th fixation

$U(t_i, x_i, y_i)$ — displacement field

GazeMorph CNN



Input: 2 channels (page and fixations) $\times$ 200 timesteps (80 sec) $\times$ 152 (page height in px) $\times$ 104 (page width in px)

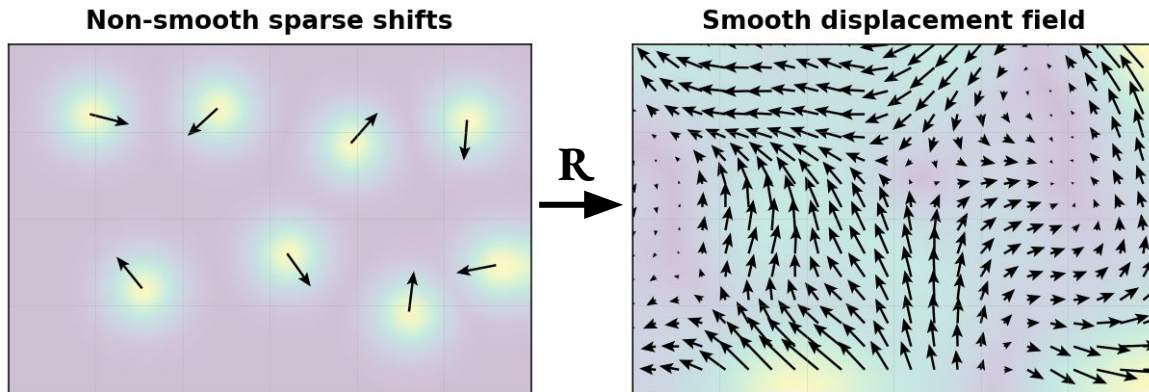
Output: 2 channels (displacements by x and y) $\times$ 200 $\times$ 152 $\times$ 104

Training Objective

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N} \sum_{i=1}^N \rho(\hat{\mathbf{f}}_i - \mathbf{f}_i)}_{\text{fixation accuracy}} + \underbrace{\lambda_s \mathbb{E}_{t,x,y} [\|\nabla_{xy} U_\theta(t, x, y)\|_2^2]}_{\text{spatial smoothness}} + \underbrace{\lambda_t \mathbb{E}_{t,x,y} [\|\partial_t U_\theta(t, x, y)\|_2^2]}_{\text{temporal smoothness}} + \underbrace{\lambda_f \mathbb{E}_{t,x,y} [\max(0, -\det J_{\text{id}+U_\theta}(t, x, y))]}_{\text{fold penalty}}$$

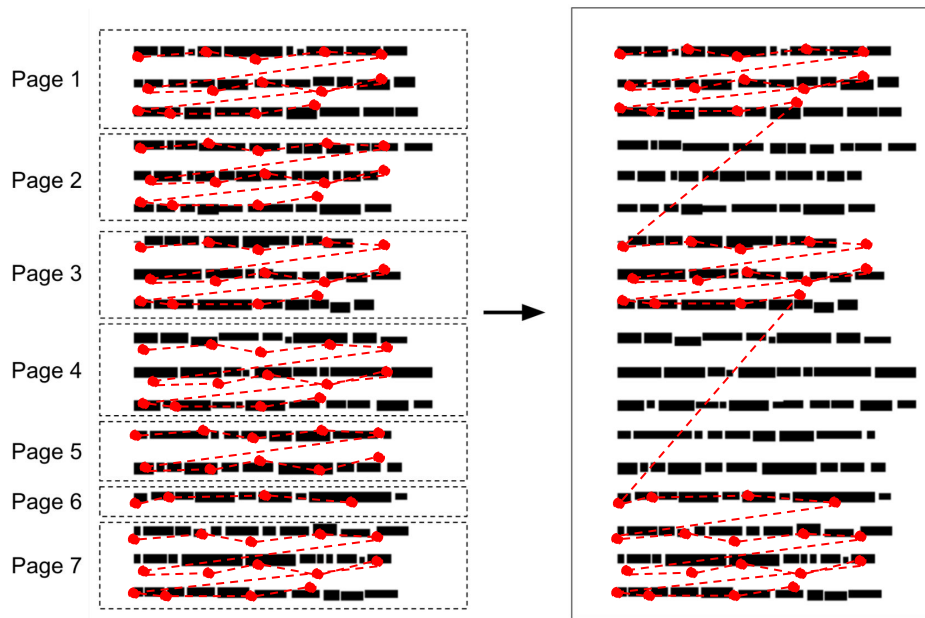
What the formula does:

1. Makes corrected fixations close to true fixations
2. Penalizes sharp spatial changes in the displacement field
3. Penalizes sudden temporal changes
4. Penalizes folded transformations

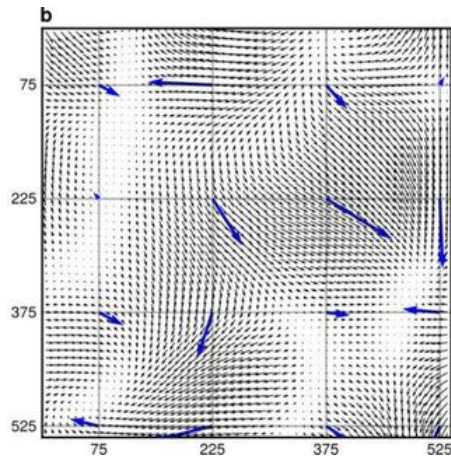


ZuCo 1.0 Dataset Preprocessing

1. Combining 7 pages from the original dataset with 50% probability of fixations transfer



2. Augmenting fixation maps using Elastic Distortion, with displacement field changing over time



$$F_x^{i+1} = F_x^i + c_i * F'_x$$

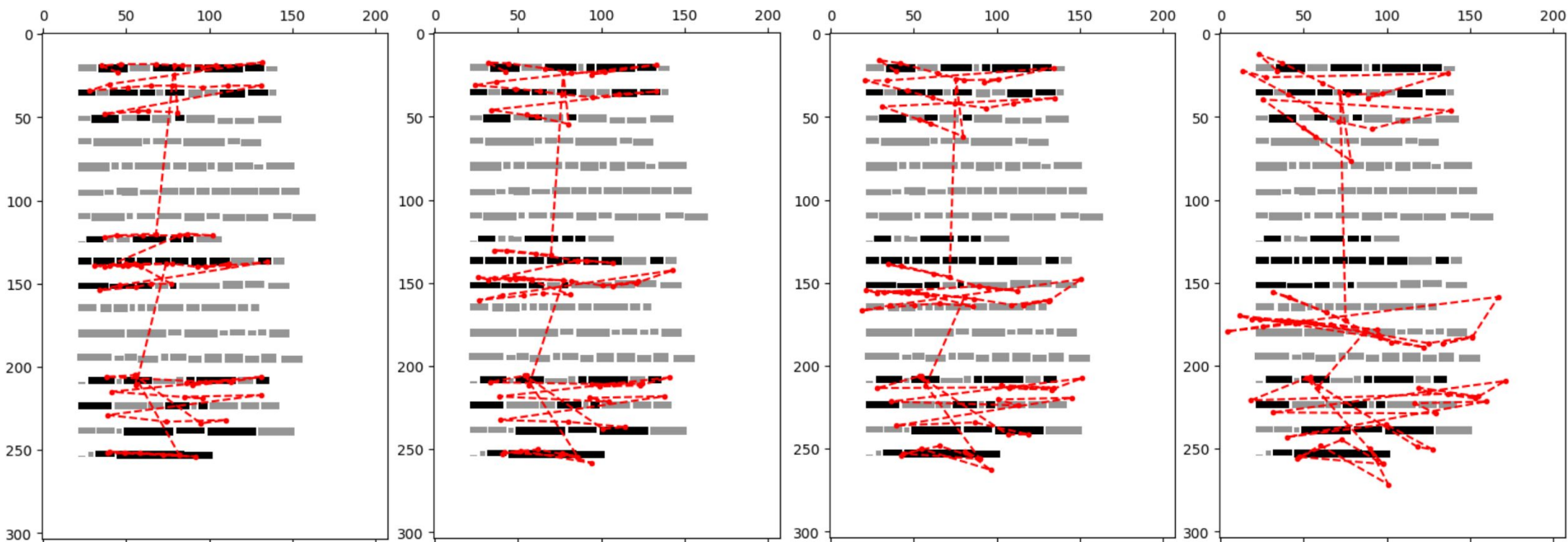
$$F_y^{i+1} = F_y^i + c_i * F'_y$$

$$c_i \in |N(0, 0.5)|^3$$

Resulting Dataset

The dataset was divided into 4 parts with different augmentation levels

Final dataset size: 18,200 pages (Train / validation / test: 14,400 / 1,600 / 2,200 pages)



(a) Without augmentation.

(b) Weak augmentation.

(c) Moderate augmentation.

(d) Strong augmentation.

Training

Optimization

- AdamW
- 25 epochs
- learning rate: $1e-4$
- weight decay: $1e-6$
- batch size: 3
- bfloat16 mixed precision
- GPU: RTX 5090 (32 Gb VRAM)

Loss Weights

$\lambda_s = 0.0375$ - spatial smoothness

$\lambda_t = 0.05$ - temporal smoothness

$\lambda_f = 0.10$ - fold penalty

Model Variants

Variant	# Params	Time
GazeMorph-S	9.8M params	~4h
GazeMorph-M	39M params	~8h
GazeMorph-L	88M params	~15h

Results

Method	<i>none</i>			<i>weak</i>			<i>moderate</i>			<i>strong</i>		
	$\bar{E}_{\text{pre}}$	$\bar{E}_{\text{post}}$	ER	$\bar{E}_{\text{pre}}$	$\bar{E}_{\text{post}}$	ER	$\bar{E}_{\text{pre}}$	$\bar{E}_{\text{post}}$	ER	$\bar{E}_{\text{pre}}$	$\bar{E}_{\text{post}}$	ER
Attach [Carr et al. 2022]	1.539	1.308	<u>0.850</u>	2.702	2.605	<u>0.964</u>	4.535	4.510	<u>0.994</u>	8.559	8.440	<u>0.986</u>
Chain [Carr et al. 2022]	1.539	1.269	<u>0.825</u>	2.702	2.487	<u>0.921</u>	4.535	4.367	<u>0.963</u>	8.559	8.349	<u>0.976</u>
Merge [Carr et al. 2022]	1.539	1.687	<u>1.096</u>	2.702	2.736	<u>1.013</u>	4.535	4.385	<u>0.967</u>	8.559	8.068	<u>0.943</u>
Cluster [Carr et al. 2022]	1.539	1.330	<u>0.864</u>	2.702	2.621	<u>0.970</u>	4.535	4.523	<u>0.997</u>	8.559	8.471	<u>0.990</u>
Stretch [Carr et al. 2022]	1.539	1.308	<u>0.850</u>	2.702	2.605	<u>0.964</u>	4.535	4.510	<u>0.994</u>	8.559	8.440	<u>0.986</u>
Regress [Carr et al. 2022]	1.539	4.657	<u>3.026</u>	2.702	5.590	<u>2.069</u>	4.535	6.867	<u>1.514</u>	8.559	10.305	<u>1.204</u>
Al-Madi [Al Madi 2025]	1.539	1.619	<u>1.052</u>	2.702	2.722	<u>1.007</u>	4.535	4.401	<u>0.971</u>	8.559	8.112	<u>0.948</u>
GazeMorph-S (9.8M)	1.539	0.949	<u>0.617</u>	2.702	1.852	<u>0.685</u>	4.535	2.895	<u>0.638</u>	8.559	4.966	<u>0.580</u>
GazeMorph-M (39M)	1.539	0.946	<u>0.615</u>	2.702	1.859	<u>0.687</u>	4.535	2.859	<u>0.630</u>	8.559	4.817	<u>0.562</u>
GazeMorph-L (88M)	1.539	0.862	<u>0.560</u>	2.702	1.802	<u>0.666</u>	4.535	2.834	<u>0.624</u>	8.559	4.828	<u>0.564</u>

$\bar{E}_{\text{pre}}$ – mean error before correction (in pixels)

$\bar{E}_{\text{post}}$ – mean error after correction (in pixels)

$$\text{ER} = \frac{\bar{E}_{\text{post}}}{\bar{E}_{\text{pre}}}$$

Ablation: Effect of Page Content

Page content improves correction across all augmentation levels.

Augmentation	ER (with content)	ER (without content)
<i>none</i>	0.617	0.758
<i>weak</i>	0.685	0.818
<i>moderate</i>	0.638	0.789
<i>strong</i>	0.580	0.686

With content: GazeMorph-S receives the page content mask.

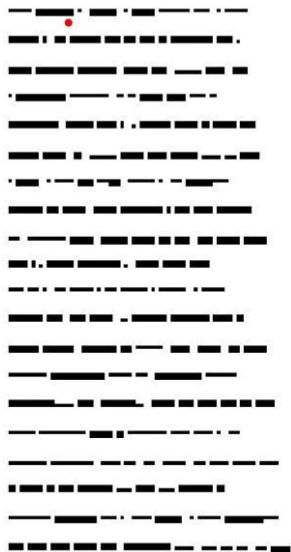
Without content: the content channel is replaced with an all-ones mask.

Conclusion

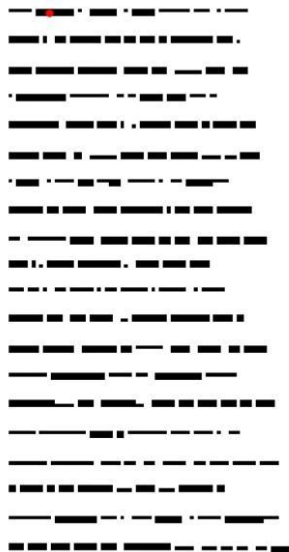
1. Proposed GazeMorph for reducing eye-tracking measurement error
2. Showed that GazeMorph outperforms non-learning baselines across different noise levels
3. Demonstrated that content structure can be used as a strong signal for fixation correction
4. Built an augmented ZuCo-based dataset with different levels of simulated measurement error

Demo

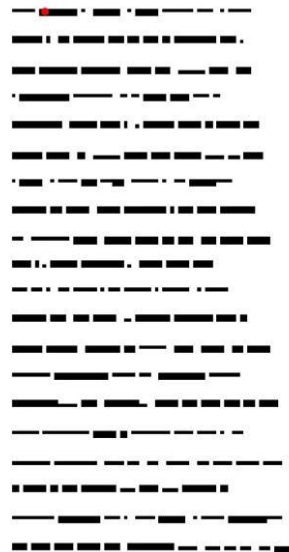
Raw



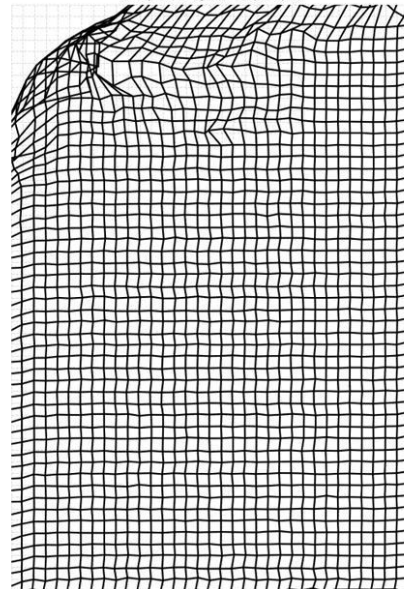
Warped



Ground-Truth



Warp field (grid; step=3px)



Thank you for attention!



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 Telegram: [@qskxx](https://t.me/@qskxx)

Future Work

1. Broader dataset – move from text pages to arbitrary visual content, such as web pages, images, and interfaces.
2. Using the raw page image instead of a mask – this can improve performance and make the model more general
3. Using the encoder-decoder model not for correction, but for gaze registration – instead of passing registered fixations, pass eye images and predict gaze position with a content prior.